Combinatorial idebtities

Most important identities:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

so $$2^n = (1+1)^n = \sum_{k=0}^{n} \binom{n}{k}$$

where $\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$ stems also from Pascal's triangle

$$\begin{array}{ll}
n=1 & 1 \\
n=2 & 1 \quad 1 \\
n=3 & 1 \quad 2 \quad 1 \\
n=4 & 1 \quad 3 \quad 3 \quad 1 \\
n=5 & 1 \quad 4 \quad 6 \quad 4 \quad 1 \\
 & \uparrow \quad \uparrow \quad \uparrow \\
 & k=1 \quad k=2 \quad k=3
\end{array}
\qquad \downarrow n \qquad \longrightarrow k$$

$$\binom{n}{k} = \binom{5}{4}$$

1) Expand

$$(1+x)^6 = 1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6$$

$$(1-x)^6 = 1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6$$

2) $\binom{13}{6} = 1716$, $\binom{17}{8} = 24310$

3) $\binom{n+1}{3} + \binom{n-1}{3} = (n-1)^2$

a) direct

$$n^2+n-(n^2-5n+6)$$

$$\binom{n+1}{3} = \frac{(n+1)\,n\,(n-1)}{6}$$

$$-\binom{n-1}{3} = \frac{(n-1)(n-2)(n-3)}{6}$$

$$\frac{(n-1)[n(n+1)-(n-2)(n-3)]}{6} = (n-1)^2$$

b) Observe that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1} \quad \text{from Pascal's triangle}$$

so

$$\binom{n+1}{3} = \binom{n}{3} + \binom{n}{2}$$

$$-\binom{n-1}{3} = \binom{n}{3} - \binom{n-1}{2}$$

$$\binom{n}{2} + \binom{n-1}{2} = (n-1)^2$$

c) combinatorial

$\binom{n+1}{3}$ - 3-element subset

$\binom{n-1}{3}$ - 3-element avoiding 2 numbers

difference - 3-element subsets not containing 2 numbers (e.g. 1, n+1)

- containing 1, but not n+1: $\binom{n-1}{2}$
- containing n+1, but not 1: $\binom{n-1}{2}$
- containing both: $n-1$

$\Big\}$ + $2\binom{n-1}{2} + n-1 = (n-1)^2$

4. • $\sum_{k=0}^{n} \binom{n}{k}\binom{n-k}{m-k} = 2^m \binom{n}{m}$

observe $\binom{n}{k}\binom{n-k}{m-k} = \binom{n}{m}\binom{m}{k}$

choose m elements from n, and from those select k

so $\sum_{k=0}^{n} \binom{n}{k}\binom{n-k}{m-k} = \sum_{k=0}^{m} \binom{n}{m}\binom{m}{k}$

because for $k>m$ $\binom{m}{k}=0$

$$= \binom{n}{m} \underbrace{\sum_{k=0}^{m} \binom{m}{k}}_{= 2^m}$$

• $\sum_{k=1}^{n} k^2 \binom{n}{k} = n(n+1)2^{n-2}$

First observe:

$$\boxed{\sum_{k=0}^{n} k \binom{n}{k} = n\, 2^{n-1}}$$

start with $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$

$\Downarrow$

$$n(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^{k-1}$$

so $n\, 2^{n-1} = \sum_{k=0}^{n} k \binom{n}{k}$ by setting $x=1$

now

$$nx(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^k$$

differentiating: $n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^{k-1}$

at $x=1$

$$n \cdot 2^{n-1} + n(n-1)2^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k}$$

$$\parallel$$

$$n\,2^{n-2}(2+n-1) = n(n+1)\,2^{n-2}$$

so $\displaystyle\sum_{k=0}^{n} k^2 \binom{n}{k} = n(n+1)\,2^{n-1}$

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also

$$\sum_{k=0}^{n} k(k-1)\binom{n}{k} = n(n-1)\,2^{n-1}$$

And observe

$$k^2 = k(k-1) + k$$

so $\displaystyle\sum_{k=0}^{n} k^2 \binom{n}{k} = \sum_{k=0}^{n} k(k-1)\binom{n}{k} + \sum_{k=0}^{n} k \binom{n}{k}$

because

$$\sum_{k=0}^{n} k \binom{n}{k} = n\,2^{n-1}$$

$$+ \sum_{k=0}^{n} k(k-1)\binom{n}{k} = n(n-1)\,2^{n-2}$$

$$n(n-1)\,2^{n-2}$$

- by combinatorics:

$k^2\binom{n}{k}$ number of ordered pairs $(x,y) \in A \times A$ where $|A| = k$

$\sum k^2\binom{n}{k} \to$ all triples (A, x, y) $A \subseteq S$, $x, y \in A$

other way $\to$ choose x, y first

$\to x \neq y \to n(n-1)$ choices, 2^{n-2} other elements may, or may not be in A

$x = y$ n choices for x, 2^{n-1} for other.

sum $n(n-1)2^{n-2} + n2^{n-1} = \sum_{k=0}^{n} k^2\binom{n}{k}$

- by probability:

$X \sim \text{Binom}(n, 1/2) \to$ number of successes in n throws.

$$P(X=k) = \frac{\binom{n}{k}}{2^n}$$

$$\mathbb{E}[X^2] = \sum_{k=0}^{n} k^2 P(X=k) = \frac{1}{2^n}\sum_{k=0}^{n} k^2\binom{n}{k}$$

Also $\mathbb{E}[X^2] = Var(X) + (\mathbb{E}[X])^2$

$\mathbb{E}[X] = n \cdot p = n \cdot \frac{1}{2}$

$Var(X) = np(1-p) = n \cdot \left(\frac{1}{2}\right)^2$

so $\mathbb{E}[X^2] = \frac{n}{2^2} + \frac{n^2}{2^2} =$ simplify and get result.

$= \frac{n}{4}(1+n)$

Also

- $k\binom{n}{k} = n\binom{n-1}{k-1}$

- $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$

$k\binom{n}{k}$ ← choose k element subset, choose one element from it.

<u>so choose (A, x) pair</u>

because

$k^2 = k(k-1) + k$ so

$$k^2\binom{n}{k} = k(k-1)\binom{n}{k} + k\binom{n}{k} =$$

$$= n(n-1)\binom{n-2}{k-2} + n\binom{n-1}{k-1}$$

now $\sum_{k=0}^{n}\binom{n-2}{k-2} = \sum_{j=0}^{n-2}\binom{n-2}{j} = 2^{n-2}$

$\sum_{k=0}^{n}\binom{n-1}{k-1} = \sum_{j=0}^{n-2}\binom{n-1}{j} = 2^{n-1}$

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5. $\sum_{k=0}^{n} 2^k\binom{n}{k}$

$$(1+x)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$$

for $x=2$ LHS $= 3^n$ so $\sum_{k=0}^{n} 2^k\binom{n}{k} = 3^n$

b) $\sum_{k=1}^{n} k\, 3^k \binom{n}{k}$

we have $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$

differentiating $n(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^{k-1}$

multiplying by x

$$n(1+x)^{n-1} x = \sum_{k=0}^{n} k \binom{n}{k} x^k$$

so $\sum_{k=1}^{n} k\, 3^k \binom{n}{k} = n(1+3)^{n-1} \cdot 3 = \underline{3n4^{n-1}}$

6.

a) $m \binom{m+n-1}{k-1} = \sum_{i=1}^{k} i \binom{m}{i} \binom{n}{k-i}$

Take sets M (men) and N (women) ← one man distinguish with red tie

choose a group of k elements with one distinguished element

$m \binom{m+n-1}{k-1}$

distinguished ↑ ↑ $k-1$ elements from the rest.

or...

chose i from M, $k-i$ from N, pick distinguished one, sum

b) $\binom{2n}{2} = 2\binom{n}{2} + n^2$

split $2n$ set to sets A, B of size n

choose 2 elements from $(A \cup B)$

both from A or both from B : $\binom{n}{2}$

one from A, one from B : n^2

c) $\sum_{i=0}^{n} \binom{n}{i} i = n 2^{n-1}$

count pairs (A, x), choose x, other elements were, or were not in A

d) $\sum_{j=0}^{n} \binom{n}{j} \binom{n-j}{k-j} = \binom{n}{k} 2^k$

$\binom{n}{j} \binom{n-j}{k-j} = \binom{n}{k} \binom{k}{j}$ → substitute and it is solved

take j elements from k

Elements of $S \times \{1, 2\}$

Inclusion exclusion

1. How many integers there are from 1-1000 divisible by none of

a) 2, 6, 13

divisible by 2 $\rightarrow \left\lfloor \frac{1000}{2} \right\rfloor = 500$

by 13 $\rightarrow \left\lfloor \frac{1000}{13} \right\rfloor = 76$

by both:

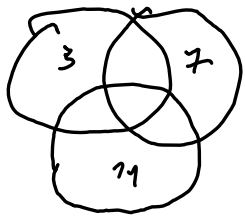
$\left\lfloor \frac{1000}{26} \right\rfloor = 38$

by inclussion - exclussion

divisible by at least one: $500 + 76 - 38 = 538$

not divisible: $1000 - 538 = 462$

b) 3, 7, 11



by 3 $\rightarrow \left\lfloor \frac{1000}{3} \right\rfloor = 333$

7 $\rightarrow \left\lfloor \frac{1000}{7} \right\rfloor = 142$

11 $\rightarrow \left\lfloor \frac{1000}{11} \right\rfloor = 90$

3, 7 $\rightarrow \left\lfloor \frac{1000}{21} \right\rfloor = 47$

3, 11 $\rightarrow \left\lfloor \frac{1000}{33} \right\rfloor = 30$

7, 11 $\rightarrow \left\lfloor \frac{1000}{77} \right\rfloor = 12$

3, 7, 11 $\rightarrow \left\lfloor \frac{1000}{231} \right\rfloor = 4$

$333 + 142 + 90 - 47 - 30 - 12 + 4 = 480$

not divisible: 520

c) 6, 63, 144

$144 = 12^2 = 6 \cdot 12 \cdot 2$ → wystarczy wziąć 6, 63

$\left\lfloor \frac{1000}{6} \right\rfloor = 166$

$\left\lfloor \frac{1000}{63} \right\rfloor = 15$

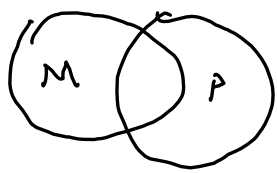
$\left\lfloor \frac{1000}{126} \right\rfloor = 7$

$166 + 15 - 7 = 174$

$1000 - 174 = 826$

2) Each of class of 50 students studies at least one from Physics or Mathematics. 30 takes math, 27 both.

How many reads physics?



$|M \cup P| = 50$

$|M| = 30$

$|M \cap P| = 27$

$50 = 30 + P - 27 \Rightarrow P = 47$